

Matris-Zinciri Çarpımı

Örnek

Matrix-Chain Multiplication

- $A_1A_2A_3A_4$.
- $(A_1(A_2(A_3A_4)))$
- $(A_1((A_2A_3)A_4))$
- $((A_1A_2)(A_3A_4))$
- $((A_1(A_2A_3))A_4)$
- $((((A_1A_2)A_3)A_4))$

$$f(n) = \sum_{k=1}^{n-1} f(k)f(n-k).$$

$$f(n) = \frac{1}{n} \binom{2n-2}{n-1}.$$

İki matrisin çarpımı

➤ **Input:** $A_{p \times q}$ ve $B_{q \times r}$ matrisleri

➤ **Output:** $A \cdot B$ matrislerinin çarpımı: $C_{p \times r}$

Algorithm Matrix-Multiply($A_{p \times q}$, $B_{q \times r}$)

```
{  for  $i := 1$  to  $p$  do
    for  $j := 1$  to  $r$  do
        {   $C[i, j] := 0$ ;
            for  $k := 1$  to  $q$  do
                 $C[i, j] := C[i, j] + A[i, k] \cdot B[k, j]$ ;
            }
        return  $C$ ;
}
```

➤ **Skaler çarpma sayısı** = pqr

Örnek

- $A_{10 \times 100}$, $B_{100 \times 5}$, and $C_{5 \times 50}$
- There are 2 ways to parenthesize
 - $((AB)C) = D_{10 \times 5} \cdot C_{5 \times 50}$
 - $AB \Rightarrow 10 \cdot 100 \cdot 5 = 5,000$ scalar multiplications,
 - $DC \Rightarrow 10 \cdot 5 \cdot 50 = 2,500$ scalar multiplications,
 - Total scalar multiplication = 7,500
 - $(A(BC)) = A_{10 \times 100} \cdot E_{100 \times 50}$
 - $BC \Rightarrow 100 \cdot 5 \cdot 50 = 25,000$ scalar multiplications,
 - $AE \Rightarrow 10 \cdot 100 \cdot 50 = 50,000$ scalar multiplications,
 - Total scalar multiplication = 75,000

Dinamik Programlama

➤ Step 2: A Recursif çözüm)

- $A_i A_{i+1} \dots A_j$ çarpımının rekürsif tanımı:

$$m[i, j] = \begin{cases} 0 & \text{if } i = j, \\ \min_{i \leq k < j} \{m[i, k] + m[k+1, j] + p_{i-1}p_kp_j\} & \text{if } i < j. \end{cases}$$

- Optimal çözümün yapılandırılması:
- $s[i, j] = k$; *optimal bölümlleme*.
- $m[i, j] = m[i, k] + m[k+1, j] + p_{i-1}p_kp_j$

➤ Step 3: Optimal maliyet

- Zincir uzunluğu=1 için hesapla, Daha sonra zincir uzunluğu $l=2, 3, \dots$ için optimal maliyeti hesapla
- bottom-up

Dinamik Programlama

- **Input:** Array $p[0 \dots n]$
- **Result:** Minimum maliyet tablosu: s

Algorithm MatrixChainOrder($p[]$, n)

```
{  
    for  $i := 1$  to  $n$  do  
         $m[i, i] := 0$ ;  
    for  $l := 2$  to  $n$  do  
        for  $i := 1$  to  $n-l+1$  do  
            {  $j := i+l-1$ ;  
               $m[i, j] := \infty$ ;  
              for  $k := i$  to  $j-1$  do  
                  {  $q := m[i, k] + m[k+1, j] + p[i-1] p[k] p[j]$ ;  
                    if ( $q < m[i, j]$ ) then  
                        {  $m[i, j] := q$ ;  
                           $s[i, j] := k$ ;  
                        }  
                  }  
            }  
        }  
    return  $m$  and  $s$   
}
```

Örnek

➤ $O(n^3)$ time ve $O(n^2)$ space.

Matrix	A_1	A_2	A_3	A_4	A_5	A_6
Dimensions	10x20	20x5	5x15	15x50	50x10	10x15

i\j	1	2	3	4	5	6
1	0					
2		0				
3			0			
4				0		
5					0	
6						0

Örnek

- Chains of length 2 are easy, as there is no minimization required, so
- $m[i, i+1] = p_{i-1}p_i p_{i+1}$
- $m[1, 2] = 10 \times 20 \times 5 = 1000$
- $m[2, 3] = 20 \times 5 \times 15 = 1500$
- $m[3, 4] = 5 \times 15 \times 50 = 3750$
- $m[4, 5] = 15 \times 50 \times 10 = 7500$
- $m[5, 6] = 50 \times 10 \times 15 = 7500$

Matrix	A_1	A_2	A_3	A_4	A_5	A_6
Dimensions	10x20	20x5	5x15	15x50	50x10	10x15

i\j	1	2	3	4	5	6
1	0	1000				
2		0	1500			
3			0	3750		
4				0	7500	
5					0	7500
6						0

Matrix	A_1	A_2	A_3	A_4	A_5	A_6
Dimensions	10x20	20x5	5x15	15x50	50x10	10x15

- $m[1,3] = \min\{(m[1,1] + m[2,3] + p_0 p_1 p_3), (m[1,2] + m[3,3] + p_0 p_2 p_3)\}$
 $= \min\{(0 + 1500 + 10 \times 20 \times 15), (1000 + 0 + 10 \times 5 \times 15)\}$
 $= \min\{4500, 1750\} = 1750$
- $m[2,4] = \min\{(m[2,2] + m[3,4] + p_1 p_2 p_4), (m[2,3] + m[4,4] + p_1 p_3 p_4)\}$
 $= \min\{(0 + 3750 + 20 \times 5 \times 50), (1500 + 0 + 20 \times 15 \times 50)\}$
 $= \min\{8750, 16500\} = 8750$
- $m[3,5] = \min\{(m[3,3] + m[4,5] + p_2 p_3 p_5), (m[3,4] + m[5,5] + p_2 p_4 p_5)\}$
 $= \min\{(0 + 7500 + 5 \times 15 \times 10), (3750 + 0 + 5 \times 50 \times 10)\}$
 $= \min\{8250, 6250\} = 6250$
- $m[4,6] = \min\{(m[4,4] + m[5,6] + p_3 p_4 p_6), (m[4,5] + m[6,6] + p_3 p_5 p_6)\}$
 $= \min\{(0 + 7500 + 15 \times 50 \times 15), (7500 + 0 + 15 \times 10 \times 15)\}$
 $= \min\{18750, 9750\} = 9750$

i\j	1	2	3	4	5	6
1	0	1000	1750			
2		0	1500	8750		
3			0	3750	6250	
4				0	7500	9750
5					0	7500
6						0

Matrix	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆
Dimensions	10x20	20x5	5x15	15x50	50x10	10x15

➤ $m[1,4] = \min\{(m[1,1] + m[2,4] + p_0 p_1 p_4), (m[1,2] + m[3,4] + p_0 p_2 p_4), (m[1,3] + m[4,4] + p_0 p_3 p_4)\}$
 $= \min\{(0 + 8750 + 10 \times 20 \times 50), (1000 + 3750 + 10 \times 5 \times 50), (1750 + 0 + 10 \times 15 \times 50)\}$
 $= \min\{18750, 7250, 9250\} = 7250$

➤ $m[2,5] = \min\{(m[2,2] + m[3,5] + p_1 p_2 p_5), (m[2,3] + m[4,5] + p_1 p_3 p_5), (m[2,4] + m[5,5] + p_1 p_4 p_5)\}$
 $= \min\{(0 + 6250 + 20 \times 5 \times 10), (1500 + 7500 + 20 \times 15 \times 10), (8750 + 0 + 20 \times 50 \times 10)\}$
 $= \min\{7250, 12000, 18750\} = 7250$

➤ $m[3,6] = \min\{(m[3,3] + m[4,6] + p_2 p_3 p_6), (m[3,4] + m[5,6] + p_2 p_4 p_6), (m[3,5] + m[6,6] + p_2 p_5 p_6)\}$
 $= \min\{(0 + 9750 + 5 \times 15 \times 15), (3750 + 7500 + 5 \times 50 \times 15), (6250 + 0 + 5 \times 10 \times 15)\}$
 $= \min\{10875, 15000, 7000\} = 7000$

i\j	1	2	3	4	5	6
1	0	1000	1750	7250		
2		0	1500	8750	7250	
3			0	3750	6250	7000
4				0	7500	9750
5					0	7500
6						0

Matrix	A_1	A_2	A_3	A_4	A_5	A_6
Dimensions	10x20	20x5	5x15	15x50	50x10	10x15

- $m[1,5] = \min\{(m[1,1]+m[2,5]+p_0p_1p_5), (m[1,2]+m[3,5]+p_0p_2p_5),$
 $(m[1,3]+m[4,5]+p_0p_3p_5), (m[1,4]+m[5,5]+p_0p_4p_5)\}$
 $= \min\{(0+7250+10 \times 20 \times 10), (1000+6250+10 \times 5 \times 10),$
 $(1750+7500+10 \times 15 \times 10), (7250+0+10 \times 50 \times 10)\}$
 $= \min \{ 9250, 7750, 10750, 12250 \} = 7750$
- $m[2,6] = \min\{(m[2,2]+m[3,6]+p_1p_2p_6), (m[2,3]+m[4,6]+p_1p_3p_6),$
 $(m[2,4]+m[5,6]+p_1p_4p_6), (m[2,5]+m[6,6]+p_1p_5p_6)\}$
 $= \min\{(0+7000+20 \times 5 \times 15), (1500+9750+20 \times 15 \times 15),$
 $(8750+7500+20 \times 50 \times 15), (7250+0+20 \times 10 \times 15)\}$
 $= \min \{ 8500, 15750, 31,250, 10250 \} = 8500$
- $m[1,6] = \min\{(m[1,1]+m[2,6]+p_0p_1p_6), (m[1,2]+m[3,6]+p_0p_2p_6),$
 $(m[1,3]+m[4,6]+p_0p_3p_6), (m[1,4]+m[5,6]+p_0p_4p_6),$
 $(m[1,5]+m[6,6]+p_0p_5p_6)\}$
 $= \min\{(11500, 8750, 13750, 22250, 9250 \} = 8750$

i\j	1	2	3	4	5	6
1	0	1000	1750	7250	7750	8750
2		0	1500	8750	7250	8500
3			0	3750	6250	7000
4				0	7500	9750
5					0	7500
6						0

Dinamik Programlama

- Zaman maliyeti: $O(n^3)$ Bellek maliyeti: $O(n^2)$
- **Step 4: Optimal çözümü yakalayalım**
 - minimum-maliyet tablosu: $s(m, n)$.
 - $s[i, j] = k$ değeri $A_i A_{i+1} \dots A_j$ için minimum parantezleme değeri olsun.
 - Aşağıdaki procedure optimal parantezlemeyi sunar.

Algorithm PrintOptimalPerens(s, i, j)

```
{  if (i=j) then
    Print "A"i;
  else
    {  Print "(";
      PrintOptimalPerens( $s, i, s[i, j]$ );
      PrintOptimalPerens( $s, s[i, j]+1, j$ );
      Print ")";
    }
}
```

- Cevap: 8750 skaler çarpma
- Peki bu değer matrislerin hangi sırada çarpılması ile elde edilebilir?
- İlk parantezleme $k=2$ olarak uygulanır. $s[1,6]=2$.

$$A = (A_1 A_2)(A_3 A_4 A_5 A_6)$$

$$s[1,6]=2.$$

$$A_{i..j} = A_{i..k} A_{(k+1)..j} \quad s[i, j] = k.$$

- Then from the values of k : optimal parantezleme noktası

Matrix	A_1	A_2	A_3	A_4	A_5	A_6
Dimensions	10x20	20x5	5x15	15x50	50x10	10x15

➤ ?

i\j	1	2	3	4	5	6
1	1	1	2	2	2	2
2		2	2	2	2	2
3			3	3	4	5
4				4	4	5
5					5	5
6						6

➤ $s[3,6] = 5$, $A_{3..6} = A_{3..5}A_{6..6}$ ve $s[3,5] = 4$,

$$A = ((A_1A_2)((A_3A_4)A_5)A_6).$$

➤ 8750 çarpma olduğu doğrulanabilir.